Definition 7.20

The **cofactor** is a signed minor. The cofactor of a_{ij} is usually denoted by A_{ij} and is defined as $A_{ij} = (-1)^{i+j} M_{ij}$.

For instance, consider the 3 × 3 matrix defined by $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Then the minors and cofactors of the elements a_{11}, a_{12}, a_{13} are given as follows :

(i) Minor of a_{11} is $M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22}a_{33} - a_{32}a_{23}$

Cofactor of a_{11} is $A_{11} = (-1)^{1+1} M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22}a_{33} - a_{32}a_{23}$

(ii) Minor of a_{12} is $M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = a_{21}a_{33} - a_{31}a_{23}$

Cofactor a_{12} is $A_{12} = (-1)^{1+2} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = -(a_{21}a_{33} - a_{31}a_{23})$

(iii) Minor of a_{13} is $M_{13} = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} = a_{21}a_{32} - a_{31}a_{22}$

Cofactor of a_{13} is $A_{13} = (-1)^{1+3} M_{13} = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} = a_{21}a_{32} - a_{31}a_{22}$.

Result 7.1 (Laplace Expansion)

For a given matrix $A = [a_{ij}]_{3\times3}$, the sum of the product of elements of the first row with their corresponding cofactors is the **determinant** of A.

That is, $|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$.

This can also be written using minors. That is, $|A| = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$.

The determinant can be computed by expanding along any row or column and it is important to note that the value in all cases remains the same. For example,

expansion along R_1 is $|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$.

along R_2 is $|A| = a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23}$.

along C_1 is $|A| = a_{11}A_{11} + a_{21}A_{21} + a_{31}A_{31}$.

Example 7.15

Compute all minors, cofactors of A and hence compute $|A|$ if $A = \begin{bmatrix} 1 & 3 & -2 \\ 4 & -5 & 6 \\ -3 & 5 & 2 \end{bmatrix}$. Also check that $|A|$ remains unaltered by expanding along any row or any column.

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